



Prof. Simone Buso

Xmas Homework #1

Exercise 1

Consider the two-stage amplifier in Fig. 1. The circuit parameters, at $T = 25^\circ\text{C}$, are the following:

$V_{CC} = 15\text{ V}$, $R_g = 10\text{ k}\Omega$, $R_B = 1.8\text{ M}\Omega$, $R_C = 10\text{ k}\Omega$, $R_E = 1.2\text{ k}\Omega$; $R_D = 6.8\text{ k}\Omega$, $R_S = 1\text{ k}\Omega$, $R_L = 100\text{ k}\Omega$, $R_F = 2.2\text{ k}\Omega$, $C_g = 10\text{ nF}$, $C_L = 1\text{ }\mu\text{F}$, $C_F = 1\text{ }\mu\text{F}$.

Q_1 : $V_{BE} = -0.7\text{ V}$; $\beta_F = 100$; $\beta_0 = 100$; $r_0 = +\infty$;

Q_2 : $V_t = 2\text{ V}$; $I_{DSS} = k_n \frac{W}{L} V_t^2 = 0.2\text{ mA}$; $r_0 = +\infty$;

$V_T = 25\text{ mV}$.

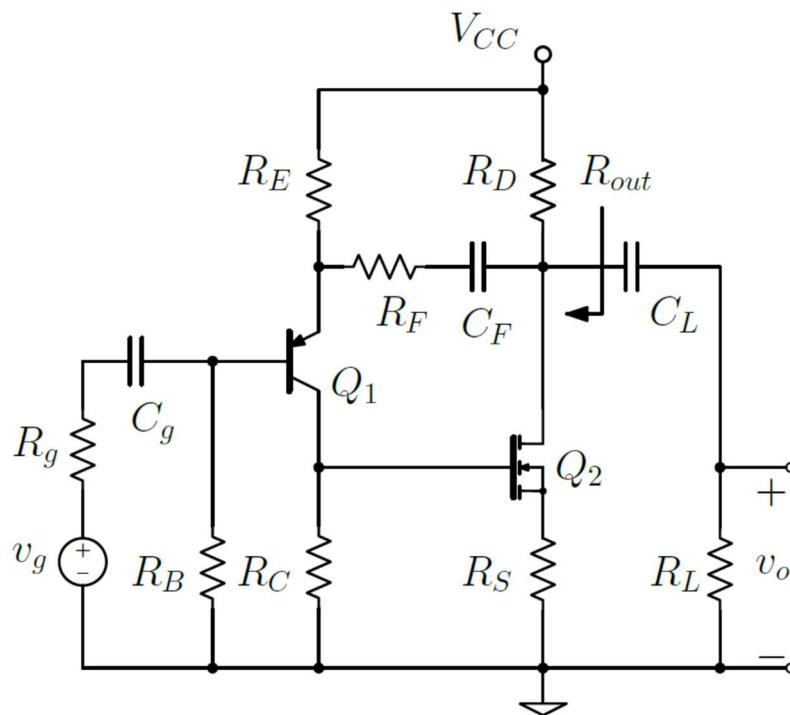


Fig. 1 – Two stage amplifier with feedback.

Considering all capacitors to be equivalent to open circuits, determine:

1. the operating points (V_{CE} , I_C) of Q_1 and (V_{DS} , I_D) of Q_2 .

Assuming all capacitors to be equivalent to *short circuits* at the frequencies of interest, determine also:

2. the voltage gain $A_v = v_o/v_g$ of the amplifier;
3. the output resistance indicated in the figure;
4. the low frequency bandwidth limit of the amplifier.

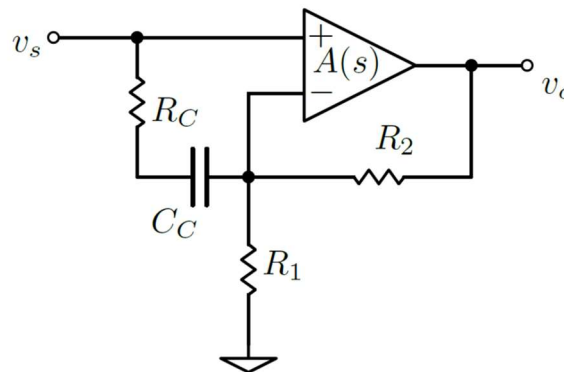


Exercise 2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 18 \text{ k}\Omega$; $R_2 = 56 \text{ k}\Omega$.

$$A(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)} \text{ [V/V], with } A_0 = 10^5 \text{ [V/V]; } \omega_{p1} = 10^2 \text{ rad/s; } \omega_{p2} = 5 \cdot 10^5 \text{ rad/s.}$$



Determine:

1. an estimation of the circuit *phase* margin with $R_C = +\infty$;
2. a block diagram representation of the amplifier;
3. the expression of the closed loop gain of the compensated amplifier, assuming $|T| \gg 1$;
4. the values of R_C and C_C that yield a $+45^\circ$ phase margin (hint: place the crossover frequency of loop gain T *exactly a decade lower than* ω_{p2} , and place the compensation network zero *at the same frequency*).